DECISION-WEIGHTED D-OPTIMAL PROBES FOR DEPTH-ONE QAOA LANDSCAPE RECOVERY

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Abstract. A reusable subspace model of parameter landscapes is useful only when a small set of observed angles determines both the surface and its optimizer. We introduce a decision-weighted D-optimal probe rule for depth-one MaxCut QAOA landscapes. Its objective adds a modular optimizer-sensitivity score to the regularized log determinant of the sampled subspace. We prove monotonicity, submodularity, the greedy approximation guarantee, and a deterministic recovery bound under model residual and observation noise. The evaluation uses eight checksum-authenticated Stanford Gset graphs and a development-only subspace learned from 120 synthetic graph landscapes. With 16 of 1024 angle values per target, the proposed design recovers six of eight grid optimizers and has maximum grid regret 1.734×10^{-3}. Unweighted D-optimal probes recover four optimizers with maximum regret 1.997×10^{-3}; 64 seeded random designs have median maximum regret 3.378×10^{-2}. The proposed design does not have the lowest mean surface error, and its advantage vanishes for some larger probe budgets. A row-Gram eigensolver is faster than the hierarchical action iteration at the evaluated dimensions, so no end-to-end speedup is claimed. The resulting contribution is a decision-aware sampling rule with an explicit validity boundary, not a universal compression claim.

Key words. QAOA, optimal design, submodular selection, response surfaces, MaxCut

1. Landscape recovery. Let $\ell_G \in \mathbb{R}^q$ collect the depth-one MaxCut QAOA expectation on a fixed 32×32 angle grid. Exact causal locality makes every entry a sum of edge-type responses [3]. A development corpus supplies a mean μ and an orthonormal basis $U \in \mathbb{R}^{q\times r}$, while a target graph supplies only the coordinates in $S \subset \{1,\dots,q\}$:

$$\ell_G = \mu + Ua + e, \qquad y_S = P_S\ell_G + \eta. \tag{1.1}$$

The coefficient estimate and recovered surface are

$$\widehat{a} = (U_S^TU_S + \lambda I)^{-1}U_S^T(y_S - \mu_S), \qquad \widehat{\ell} = \mu + U\widehat{a}. \tag{1.2}$$

The scientific loss is reported twice: relative surface error and grid regret $[\max_j \ell_j - \ell_{\arg\max_j \widehat{\ell}_j}]/\max_j \ell_j$.

2. Decision-weighted D-optimal selection. Write u_i^T for row i of U and let $w_i \ge 0$ be an optimizer-sensitivity score computed only from development landscapes. For $\alpha \ge 0$, define

$$f(S) = \log\det\left(\lambda I + \sum_{i\in S} u_iu_i^T\right) - r\log\lambda + \alpha\sum_{i\in S} w_i. \tag{2.1}$$

At every step the algorithm appends the angle with the largest marginal gain. The inverse information matrix is updated by Sherman–Morrison.

Theorem 1 (selection guarantee and cost)**.** *For $\lambda > 0$, $\alpha \ge 0$, and $w_i \ge 0$, the set function in Eq. (2.1) is normalized, nonnegative, monotone, and submodular. For a budget k, greedy selection returns S_k satisfying*

$$f(S_k) \ge (1 - e^{-1}) \max_{|S|\le k} f(S). \tag{2.2}$$

The implemented dense-basis selection requires $O(kqr^2)$ arithmetic and $O(qr + r^2)$ storage; it never forms a $q \times q$ covariance matrix.

Proof. Let $M_S = \lambda I + \sum_{i\in S} u_iu_i^T$. The matrix determinant lemma gives the marginal gain

$$\Delta_j(S) = \log(1 + u_j^TM_S^{-1}u_j) + \alpha w_j \ge 0. \tag{2.3}$$

If $S \subseteq T$, then $M_S \preceq M_T$ and $M_S^{-1} \succeq M_T^{-1}$, hence $\Delta_j(S) \ge \Delta_j(T)$. This proves monotonicity and diminishing returns; subtraction of $r\log\lambda$ gives $f(\varnothing) = 0$. The cardinality-constrained greedy bound then follows from the standard monotone-submodular recurrence [1, 2]. Each of k iterations evaluates q quadratic forms of dimension r and performs one rank-one inverse update, which proves the resource count. □

Table 1: Recovery from 16 of 1024 angle values on eight authenticated Gset targets. The interval is the interquartile range across 64 fixed random seeds. The last column is $\sigma_{\min}(U_S)$.

Probe rule	Median rel. error	Random IQR	Max. regret	Exact optima	$\sigma_{\min}$
random (64 seeds)	2.467e-02	[1.55e-02, 3.78e-02]	3.378e-02	2.0/8	2.666e-03
Morton-spaced	3.682e-02	–	1.746e-02	4/8	1.215e-04
largest leverage	4.392e-02	–	3.473e-02	4/8	6.457e-07
unweighted log-det	4.061e-03	–	1.997e-03	4/8	1.260e-01
decision-weighted log-det	4.041e-03	–	1.734e-03	6/8	1.305e-01

The theorem guarantees the declared design objective, not grid regret. The weight α is therefore calibrated on 40 development-only validation landscapes with the loss

$$\mathcal{L}_{\text{dev}} = \text{mean(relative error)} + 100\,\text{mean(grid regret)}. \tag{2.4}$$

The smallest numerically tied α is selected before public targets are opened.

Theorem 2 (deterministic recovery certificate)**.** *Let $\sigma = \sigma_{\min}(U_S)$ and let a and e be as in Eq.* (1.1)*. The ridge reconstruction in Eq.* (1.2) *satisfies*

$$\left\|\widehat{\ell} - \ell\right\|_\infty \le \|e\|_\infty + \frac{\lambda\,\|a\|_2 + \|U_S\|_2\,(\|e_S\|_2 + \|\eta\|_2)}{\sigma^2 + \lambda}. \tag{2.5}$$

Consequently the unnormalized grid regret is at most twice the right-hand side of Eq. (2.5)*.*

Proof. Subtracting a from Eq. (1.2) gives

$$\widehat{a} - a = (U_S^T U_S + \lambda I)^{-1}[U_S^T(e_S + \eta) - \lambda a]. \tag{2.6}$$

The inverse norm is $(\sigma^2 + \lambda)^{-1}$ and $\|U(\widehat{a} - a)\|_\infty \le \|\widehat{a} - a\|_2$ because every row of an orthonormal-column matrix has norm at most one. The triangle inequality proves Eq. (2.5). If j^* and $\widehat{j}$ maximize ℓ and $\widehat{\ell}$, respectively, then

$$\begin{aligned} \ell_{j^*} - \ell_{\widehat{j}} &\le |\ell_{j^*} - \widehat{\ell}_{j^*}| + |\widehat{\ell}_{\widehat{j}} - \ell_{\widehat{j}}| \\ &\le 2\left\|\widehat{\ell} - \ell\right\|_\infty. \end{aligned} \tag{2.7}$$

□

3. Authenticated evaluation. The public targets are G1, G6, G11, G14, G18, G22, G32, and G43 from the Stanford Gset collection [4]. Their files contain 800–2000 vertices and 1600–19990 edges and are verified against registered SHA-256 hashes before parsing. Edge weights are intentionally discarded, so the estimand is unweighted MaxCut. The $q = 1024$ target values are computed from the exact depth-one causal formula and independently checked on small graphs against a statevector evaluator.

The development matrix has 120 rows. A centered rank-12 right subspace is used for every target, and the audit budget is 16 observed angle values. The random comparator comprises 64 fixed seeds at every budget. Morton-spaced, largest-leverage, unweighted D-optimal, and decision-weighted D-optimal designs use the identical basis and recovery rule.

Table 1 shows that decision weighting changes the estimand rather than uniformly lowering surface error. Its mean relative error is slightly larger than that of unweighted D-optimal selection, while its maximum regret is 13.2% lower and it recovers two additional grid optimizers. The 64-seed random median has 6.1 times larger median surface error and 19.5 times larger maximum regret. Figure 1 resolves these aggregate comparisons by probe budget and by confirmation graph.

At 24 and more probes, the selected decision weight is often zero and the two D-optimal designs coincide. The method is therefore most consequential in the low-budget regime; it is not a blanket replacement for unweighted design.

Figure 2 evaluates Theorem 2 under five noise levels. All simulated errors remain below the deterministic bound, but the bound is conservative. The runtime panel supplies a separate negative control: at rank 12, row-Gram learning takes a median 2.98 ms, hierarchical actions 8.36 ms, and thin SVD 27.10 ms. Reduced retained-operator storage does not imply a wall-clock advantage at 120×1024.

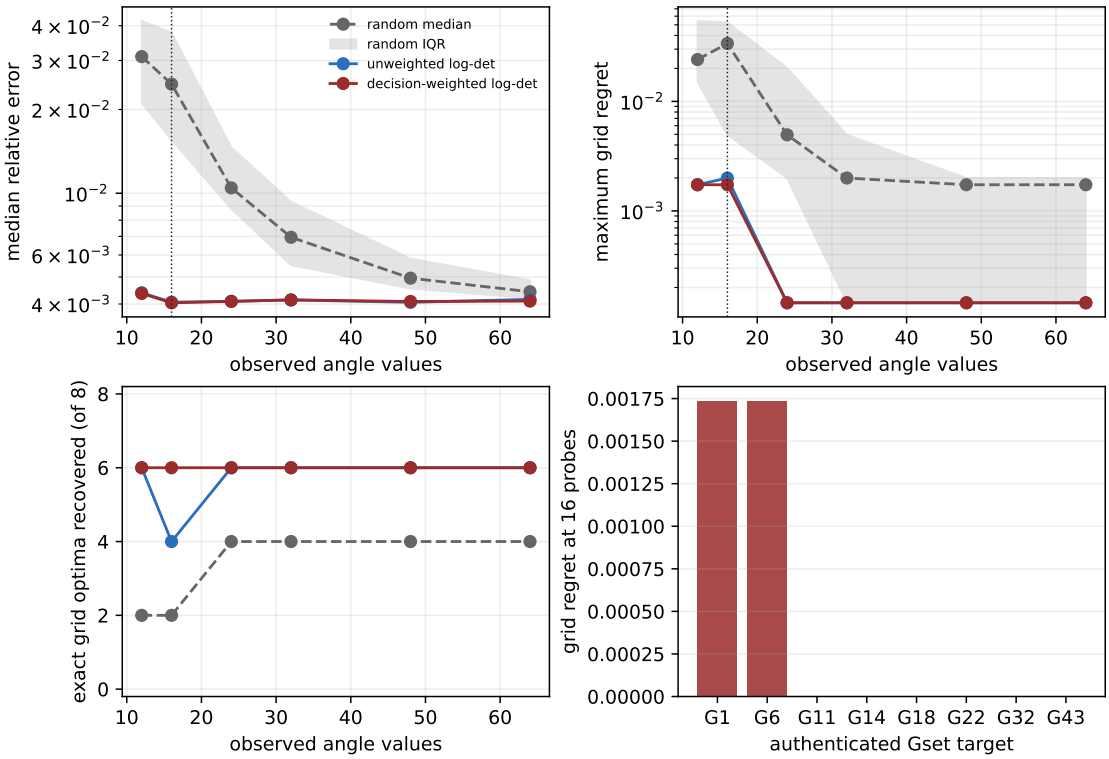


Fig. 1: Probe-budget audit. Top: surface error and grid regret, with the random interquartile range. Bottom left: exact optimizer recovery. Bottom right: targetwise regret at the predeclared 16-probe budget. The remaining regret is concentrated on G1 and G6, which share the same unweighted topology.

4. Scope and reproducibility. The package records source addresses and hashes, the complete calibration grid, all probe indices, 64 random seeds, targetwise outcomes, noise draws, operation counts, runtime repetitions, and theorem checks. A deterministic CPU command recreates the 120 development landscapes and eight public targets, regenerates the displayed artifacts, and compares the scientific payload with the locked result.

The experiment does not evaluate weighted Gset objectives, depth greater than one, hardware noise, or global MaxCut approximation quality. The method improves a low-budget angle-selection decision on this corpus. The disclosed source and budget boundaries are part of that claim.

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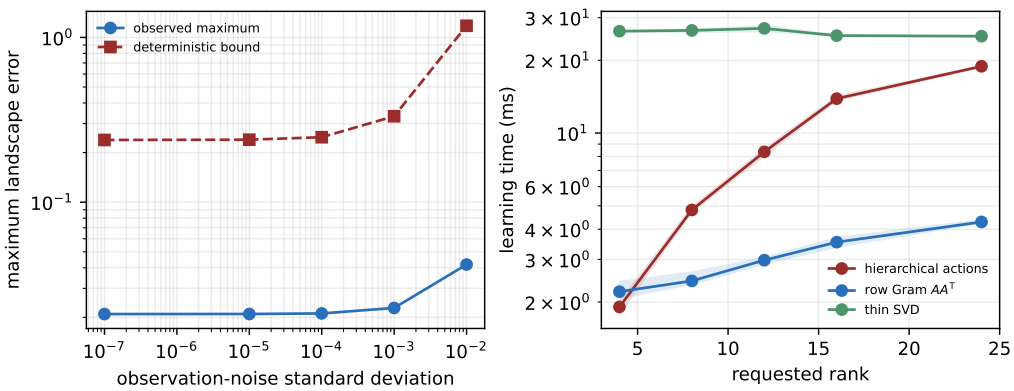


Fig. 2: Validity and cost. Left: observed maximum landscape error and the deterministic bound in Theorem 2. Right: median learning time with interquartile bands over 15 repetitions; the hierarchy construction and data-loading stages are excluded for every method.